NGUYEN THANH PHAT

Senior Full-Stack Developer

PHP · Node.js · TypeScript · React

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TECHNICAL SKILLS

Languages
PHP, TypeScript, JavaScript

Backend
PHP/Phalcon, Node.js, NestJS, REST APIs

Frontend
React, Next.js, Tailwind CSS

Databases
PostgreSQL, MongoDB, MariaDB/MySQL, Redis

ORM & APIs
TypeORM, Prisma, Drizzle, Mongoose, OpenAPI/Swagger, Socket.IO

Messaging
RabbitMQ, Apache Kafka, BullMQ

Identity & Security
Keycloak, SSO, JWT, OAuth 2.0/OIDC, RBAC, SpiceDB/ReBAC

DevOps
Linux, Docker, Docker Compose, Nginx, Apache, Git, GitLab CI/CD, PM2

Testing
Vitest, Jest, Playwright

DOMAINS

Network device management
Identity & access management
File storage & sharing
IoT & device integration
Face recognition & attendance

EDUCATION

Engineer's Degree
Computer Networks & Communications
University of Information Technology — VNU-HCM
Graduated 2020

SUMMARY

Senior Full-Stack Developer with **10+ years** building and running production systems end to end in PHP/Phalcon and Node.js/TypeScript (NestJS, Next.js). Built a 24/7 platform managing **250,000+ devices** at **~5,000 events/minute** and led the PHP → NestJS rewrite of a platform now serving **5,000+ active users**. Hands-on from architecture to deployment and production support; mentor and coordinate a team of three developers.

EXPERIENCE

HONEYNET Joint Stock Company

03/2016 – Present

Technology company researching and developing infrastructure (Wi-Fi, IoT, OS) and core solutions for digital transformation.

Senior Full-Stack Developer

2022 – Present

- Led the full rewrite of a PHP/Phalcon file-storage and identity platform to **NestJS + Next.js**, including architecture and database migration; the new platform serves **5,000+ active users**.
- Designed and scaled a 24/7 TR-069 platform managing **250,000+ devices** (~200,000 concurrently online), re-architecting background processing from process-per-request to **RabbitMQ + 20+ workers** to sustain **~5,000 events/minute** without file-descriptor, connection-pool or CPU/memory exhaustion.
- Optimized device queries by limiting retrieved fields and adding targeted indexes, reducing response times from **30+ seconds to ~5 seconds** (over 6× faster).
- Mentored and coordinated three developers while remaining hands-on across architecture, implementation, code review, deployment, troubleshooting and production support.

Full-Stack Developer

03/2016 – 2021

- Designed and built a Wi-Fi controller from the ground up with PHP/Phalcon and MariaDB for device management, RADIUS, monitoring, reporting and integration APIs; maintained and extended it in production for **8 years**.
- Built an internal operations platform covering COVID-19 declarations, QR codes, executive scheduling, task management, Face ID integration and attendance reporting; later upgraded core services from PHP 7 to PHP 8.

SOFT SKILLS

Team coordination
3 developers: tasks, estimates, technical decisions

Mentoring & coaching
Code quality, debugging, Git, security

Planning & delivery
Scope, milestones, progress tracking

Ownership & accountability
Architecture through 24/7 production support

Problem solving under pressure
Incidents: log analysis, rollback, performance fixes

Collaborative decision-making
Propose, discuss, adopt better ideas

Cross-functional communication
PM, QA and data-consuming departments

Remote collaboration
Daily updates, Git workflow, own deadlines

LANGUAGES

Vietnamese: Native
English: Technical reading & written communication

SELECTED PROJECTS

File Storage, Sharing & Identity Platform
11/2025 – Present

PHP/Phalcon → NestJS, Next.js, PostgreSQL, SpiceDB, Keycloak, MinIO, Kafka

- Implemented PostgreSQL ltree folder hierarchies, trash, public links and owner/editor/viewer permissions backed by SpiceDB.
- Implemented stateless SSO across multiple applications by verifying RS256 JWTs against Keycloak JWKS and syncing user roles from token claims.
- Built a Keycloak administration backend with bulk Excel user import/update, review workflows, audit logging and email tracking/resend.
- Integrated MinIO and Google Drive storage providers with multipart uploads, syncing MinIO bucket events through Kafka.
- Separated HTTP, worker and cron workloads; used Kafka for cross-application events and built an iframe-based File Picker SDK.

ACS Admin — TR-069 Device Management
04/2024 – 09/2026

PHP 8.3, Phalcon, MongoDB, GenieACS, RabbitMQ

- Built bulk configuration, diagnostics, network maps and condition-based Telegram/email alerts, **enabling ~5 operators to manage 250,000+ devices**.
- Implemented message identifiers to prevent duplicate processing and used database transactions to maintain consistency across multi-step updates.
- Provided integration APIs, secured with secret keys and IP allowlists, used by other departments and systems to consume device data for operational reporting.

Digital Suite for a Ward-Level Government Agency
07/2026 – 09/2026

NestJS, TypeORM, Socket.IO, Next.js, Keycloak

- Built an IAM service with bidirectional Keycloak synchronization and a Directory API for other applications.
- Built a paperless meeting system covering schedules, documents, attendees, live voting and controlled impersonation.
- Built a KPI module covering task tracking, KPI criteria and periods, and weekly reports.
- Implemented automated testing across three backend systems using Vitest and Playwright.

Face ID — Access Control & Attendance
07/2022 – 07/2026

PHP/Phalcon, MongoDB, RabbitMQ, Redis, Firebase

- Built a platform managing face-recognition devices, control commands, door locks, snapshots, people/groups and attendance reporting.
- Developed a multi-tenant version with event triggers, alert conditions, recognition schedules and worker-based bulk processing.
- Built APIs for recognition devices, plus Swagger-documented, JWT-secured APIs for the multi-tenant version.